Hilda Adwubi Osei

hildaosei109@gmail.com | website | google scholar | linkedin | github

Research Interests

- Themes: Robustness of fairness and safety guarantees in optimization and decision-making systems under imperfect information.
- Methods: Constrained optimization and reinforcement learning, applied machine learning, statistical and predictive modeling.
- Applications: AI safety, algorithmic fairness in decision-making, AI and society.

Education

Kwame Nkrumah University of Science and Technology — Kumasi, Ghana
B.Sc. Industrial Engineering — *2021 – 2024*

Publications

Work in Progress

WP1. Osei, H.A. *FarmDirect: Data-Driven pricing and risk-adjusted allocation for Ghanaian tomato.* Preprint available on ResearchGate, 2026; currently extending the multi-agent analysis with a subgroup fairness evaluation of farmer outcomes across market-volatility clusters, targeting journal submission.

Conference and Poster Presentations

CP1. Osei, C.T., Opoku Asare, A., Oti-Agyen, Y., Osei, H.A., Amooba, P. *From Knowledge to Practice: Evidence for Strengthening Infection Prevention Behaviors among Undergraduate Nursing and Midwifery Students in Ghana.* Oral presentation for Beyond Boundaries: Interdisciplinary Innovation in Health Research and Practice Virtual Conference, hosted by Johns Hopkins School of Nursing, October 19-20.

CP2. Osei, C.T., Asare, A.O., Agyen, Y.O., Osei, H.A., Amooba, P.A. *Knowledge and Compliance with Standard Precautions for Nosocomial Infection Prevention among Undergraduate Nursing and Midwifery Students at a Ghanaian University.* Poster presentation, Korle Bu Teaching Hospital, Ghana, 2026.

Preprints and Manuscripts Under Review

PM1. Osei, H.A. *Specification Gaming and a Constrained Reinforcement Learning Mitigation in a Cleaning Grid-world.* arXiv:2610.22152 [cs.LG] (cross-listed cs.AI), 2026. Under Review, Transactions of Machine Learning Research(TMLR).

PM2. Osei, H.A. *Fairness Under Prediction Error in Predict-Then-Optimize Pipelines.* arXiv:2610.18593 [cs.LG] (cross-listed cs.CY), 2026. Under Review, Transactions of Machine Learning Research(TMLR).

PM3. Osei, H.A., Osei, C.T., Asobayire, D.Y. *Learning from waste: Machine Learning for health risk prediction and computer vision-based sorting in Ghana.* arXiv:2608.25759 [cs.LG] (cross-listed cs.CV, cs.CY), 2026. Under Review, Environmental Management: Smart Solutions.

PM4. Osei, C.T., Asare, A.O., Agyen, Y.O., Osei, H.A., Amooba, P.A. *Knowledge and Compliance with Standard Precautions for Nosocomial Infection Prevention among Undergraduate Nursing and Midwifery Students at a Ghanaian University.* medRxiv, 2026. Under Review, Nursing Research and Practice.

Research and Work Experience

FarmDirect — Accra, Ghana
Founder & Lead Developer — *2025 – Present*

- Designing a data-driven marketplace to connect farmers directly to buyers, eliminating middlemen and optimizing prices; conducted the underlying research through predictive modeling, risk-adjusted optimization, and agent-based simulation on real government market data.
- Selected as 1 of 9 finalists to pitch before a national judging panel, winning an Innovation Grant (GHS 10,000); presented at the "We, The Youth" Global Youth Summit (South Korea, 1 of 150 selected from 27,000+ applicants).

Ghana Atomic Energy Commission (GAEC) Accra, Ghana
Research & Innovation Collaborator *Jun 2025 – Present*

- Conducted applied research on data-driven irrigation optimization, analyzing water-use and crop-yield data to investigate opportunities for improving agricultural resource efficiency.
- Cleaned and analyzed over 3,000 time-series data points related to water consumption and crop yields.
- Developed Python-based data visualizations and dashboards to communicate trends and support interpretation of agricultural data.
- Contributed to internal technical reports translating analytical findings into insights for agricultural decision-making and policy discussions.

Headstarter AI Remote
Software Engineering Fellow *Jul – Sep 2024*

- Built and deployed 5 AI-powered applications in 5 weeks, including a RAG-based support agent (OpenAI + Pinecone) and an LLM-driven SaaS product (Llama 3.1 via Groq).
- Applied agile, CI/CD, and product-scaling practices under mentorship from engineers at Google, Y Combinator, and Stanford.

Teaching Experience

Independent Technology Education Initiative Ghana
Programming Mentor *2026*

- Design and deliver project-based programming curriculum for early-career learners, emphasizing applied problem-solving over rote instruction.
- Guide mentees through independent coding projects, building their debugging skills and technical confidence.

National Software Development Summer School, CTVET/GAEC Ghana
Teaching Assistant *2025*

- Deliver peer-led tutorials on full-stack development (Django, Flask) to a national cohort of aspiring developers.
- Mentor participants on project architecture, version control, and pitch preparation for national innovation competitions.

MATH 152: Calculus with Analysis, Kwame Nkrumah University of Science and Technology Kumasi, Ghana
Teaching Assistant *2024*

- Led recitation sessions translating abstract analytical concepts into problem-solving strategies for undergraduates.
- Provided individualized feedback through grading and office hours to strengthen students' mathematical reasoning.

ME 259: Applications of Computer Graphics, Kwame Nkrumah University of Science and Technology Kumasi, Ghana
Teaching Assistant *2022*

- Supported students in applying computational graphics principles to engineering design problems through lab instruction.
- Reviewed and gave technical feedback on coding assignments to reinforce correct implementation practices.

Honors and Awards

- IFWY Global Explorer, International "We, The Youth" Forum, South Korea, 2025.
- Innovation & Sustainability Finalist, Mastercard Foundation x GAEC Challenge, 2025.
- STEM Women Ghana Changemaker Award, 2024.
- Academic Distinction, Second Class Upper, KNUST, 2024.
- TechWomen Africa Rising Recognition, 2023.
- Youth Arise Leadership Fellowship & Counselor Appointment, 2020.

Invited Talks

- Economic Transformation and Youth Inclusion. International "We, The Youth" Forum, UNRISD, Botswana, South Korea, 2025.
- Youth Engagement in Climate-Induced Loss and Damage in Agrifood Systems. World Food Forum, 2025.